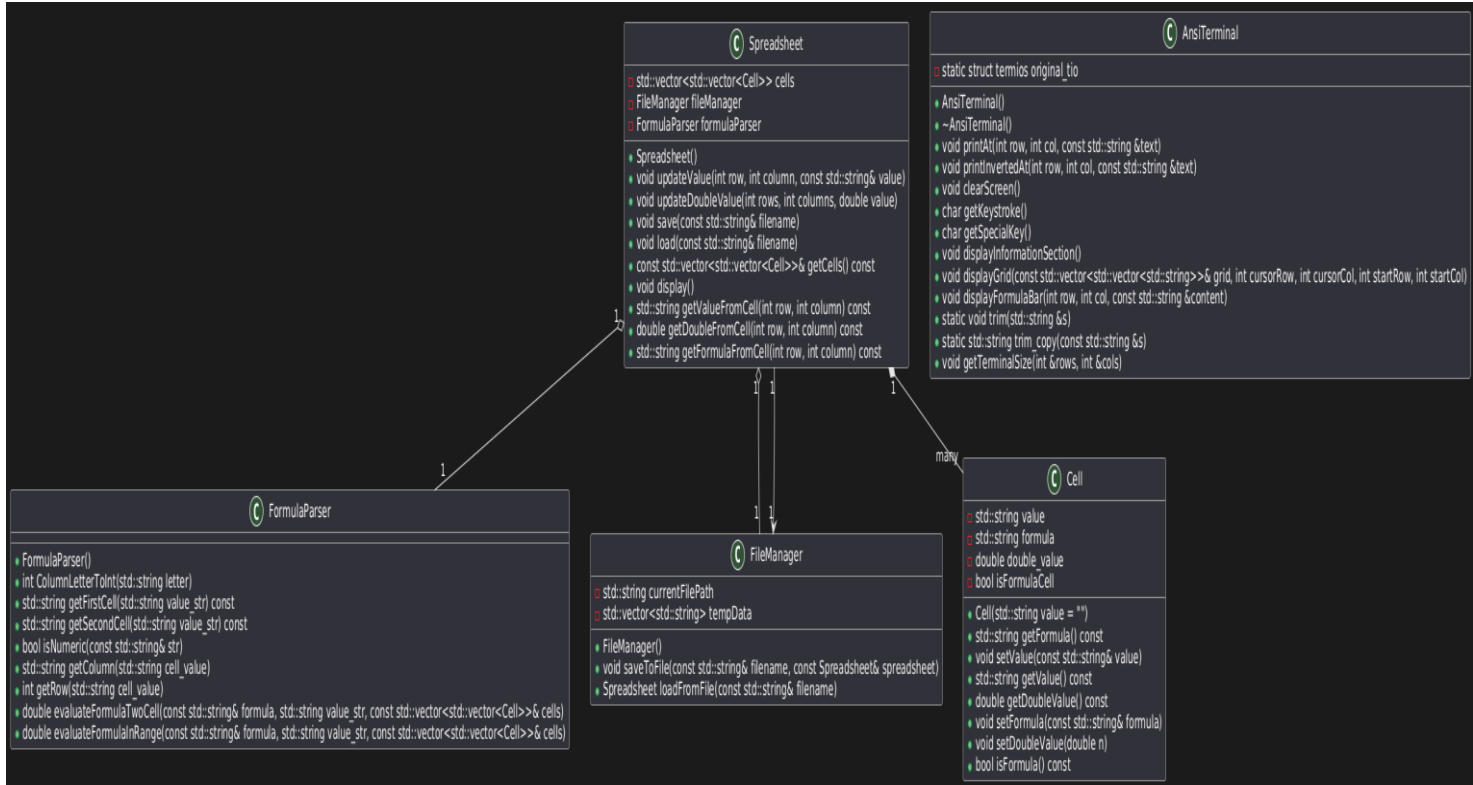


GTU CSE241 HW-2

ANSI Terminal-Based Spreadsheet Program

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Introduction

The primary objective of this project is to design and implement a simplified spreadsheet application inspired by the classic VisiCalc software. This program operates entirely within an ANSI terminal and integrates core object-oriented programming principles, such as encapsulation, inheritance, and polymorphism. The goal is to develop a functional and user-friendly application that supports basic spreadsheet operations, cell referencing, and formulas.

This project addresses several computational and structural challenges, including:

1. **Dynamic Data Storage:**

2. Utilizing a two-dimensional `std::vector` to efficiently store and manage cell data, including numbers, strings, and formulas.

3. **Formula Parsing and Evaluation:**

Implementing a robust parser to process and evaluate user-entered formulas, enabling basic arithmetic operations and cell references.

4. **Automatic Recalculation:**

Ensuring that changes to cell values automatically propagate to dependent cells, reflecting updates in real-time.

5. **File Operations:**

Supporting saving and loading spreadsheet data in CSV format to ensure compatibility with other spreadsheet applications.

6. User Interface Design:

Creating a terminal-based user interface that mimics the layout and behavior of early spreadsheet software, allowing intuitive navigation and interaction.

7.Namespaces:

Using appropriate namespaces where necessary improved the code's functionality/performance

8. Error Handling:

The code structure was improved and errors were eliminated by using try-catch

Methodology

The development process followed a structured approach:

- **Design Phase:** A UML diagram was created to map out the relationships between the classes, including Spreadsheet, Cell, FormulaParser, FileManager, and AnsiTerminal.
- **Implementation Phase:** The project was divided into modular components, with each class encapsulating a specific functionality, such as formula parsing or file management.
- **Testing and Debugging:** Several test cases, including predefined CSV files and user commands, were executed to ensure correctness and usability.

Objective

This project demonstrates the integration of C++ fundamentals, Standard Template Library (STL) features, and modern C++11 techniques. It serves as a practical application of theoretical concepts, bridging the gap between object-oriented design and real-world problem-solving. Through this project, the aim is to provide users with a simple yet powerful tool for performing spreadsheet operations in a terminal-based environment.

Features Implemented:

Cell Referencing and Automatic Recalculation:

When the value of a cell is updated, all dependent formulas are automatically recalculated, and their new values are displayed.

Advanced Formula Functions:

Functions such as SUM, AVER, STD, MIN, and MAX are fully functional.

These functions can calculate values from left to right, right to left, and even across rectangular areas.

File Operations:

The program can load data from a .csv file.

It can save the spreadsheet data into a .csv file format, ensuring compatibility with other spreadsheet applications.

Basic Arithmetic Operations:

The program supports addition, subtraction, multiplication, and division operations in formulas.

Row and Column Labeling:

Row and column labels are implemented seamlessly, allowing users to reference cells using labels like A1, B1, C3, etc.

Extended labeling for columns beyond Z (e.g., AA, BB, etc.) has also been implemented to support wider spreadsheets.

Data Storage:

All spreadsheet data is efficiently stored in a two-dimensional `std::vector` of Cell objects, ensuring smooth performance.

User Interface:

A clean and user-friendly interface is provided, with contrasting colors to enhance readability and usability in the terminal.

Missing Features

Limited Arithmetic Operations:

Arithmetic operations are limited to two operands. For example, formulas like $A1 + B2 * C1$ cannot be calculated.

Number and Cell Combination Issues:

Operations involving a combination of numbers and cell references (e.g., $A1 * 4$) are not supported. Both operands must be cell references.



User Manual

Welcome to the ANSI Terminal-Based Spreadsheet Program! This guide will walk you through the usage and features of the program to help you get started.

1. Navigating the Spreadsheet

Use the arrow keys on your keyboard to move between cells:

Up Arrow: Move to the cell above.

Down Arrow: Move to the cell below.

Left Arrow: Move to the cell to the left.

Right Arrow: Move to the cell to the right.

The selected cell is highlighted, and its label (e.g., A1, B3, etc.) is displayed at the top.

2. Selecting a Cell

Press Enter to select the currently highlighted cell.

Once a cell is selected, you can:

Enter a value (e.g., a number or text).

Enter a formula (e.g., =A1+B1 or =SUM(A1..A5)).,

3. Performing Calculations

The program supports basic arithmetic operations and advanced functions:

Arithmetic Operations:

Add (+): =A1+B1

Subtract (-): =A1-B1

Multiply (*): =A1*B1

Divide (/): =A1/B1

Advanced Functions:

These functions can calculate values across ranges:

SUM(A1..A5) - Calculates the sum of cells from A1 to A5.

AVER(A1..A5) - Calculates the average of cells from A1 to A5.

MAX(A1..A5) - Finds the maximum value in the range.

MIN(A1..A5) - Finds the minimum value in the range.

STD(A1..A5) - Calculates the standard deviation of the range.

4. Saving Your Work

Press 's' to save the current spreadsheet to a file in .csv format.

Example: If your spreadsheet contains data in A1, B1, and C1, saving the file will create a .csv file with those values.

5. Loading Data

Press L to load data from an existing .csv file into the spreadsheet.

Ensure the file format matches the expected CSV structure.

6. Exiting the Program

Press Q to exit the program.

7. User Interface Features

The top line displays the content of the currently selected cell. It also serves as the input area for entering values or formulas.

The grid layout uses labels similar to Excel:

Columns are labeled alphabetically (A, B, ..., Z, AA, AB, etc.).

Rows are labeled numerically (1, 2, 3, etc.).

The menu on the right provides a quick reference for the available commands.

Example Usage

Select cell A1 using the arrow keys and press Enter.

Enter the value 2 and press Enter again to save the value in the cell.

Move to cell B1 and enter the value 4.

Move to cell C1 and enter the formula =A1+B1.

The result, 6, will be automatically calculated and displayed in C1.

Save your work by pressing S.

Load a previously saved file by pressing **L` .

As you can see =SUM(A1..C1) equals to 12.000

Cell [D,1]: =SUM(A1..C1)

	A	B	C	D	E	F	G	H	I	J
1	2	4	6.000000	12.000000						
2										
3										
4										
5										
6										
7										
8										
9										
10										

Welcome Excell Simulation!

- 1.) Move with Arrow Keys. Select cell for press Enter
- 2.) You can calculate two cells. =A1+B1
- 3.) You can select multiple cells. =AVER(A1..C4)
- 4.) Save to file press 's'
- 5.) Load from file press 'l'
- 6.) Exit the program for press 'q'

Example .csv format after save.

```

excel.csv
1  13,13,13,13,,,,,
2  13,13,13,13,,,,,
3  13,13,13,13,13,,,,
4  13,13,13,13,13,13,,
5  ,,26.000000,,24,,
6  ,=SUM(A1..B1),,,,,
7  ,,,,,,
8  ,,,,,,
9  ,,,,,,
10 ,,,,,,
11

```

E1 is sum is rectangle. E2 is average rectangle. E3 is STDDEV of rectangle

Cell [B,2]: 50

	A	B	C	D	E	F	G	H	I	J
1	1	2	3		90.000000					
2	4	50	6		10.000000					
3	7	8	9		14.375906					
4										
5										
6										
7										
8										
9										
10										

Welcome Excell Simulation!

- 1.) Move with Arrow Keys. Select cell for press Enter
- 2.) You can calculate two cells. =A1+B1
- 3.) You can select multiple cells. =AVER(A1..C4)
- 4.) Save to file press 's'
- 5.) Load from file press 'l'
- 6.) Exit the program for press 'q'

Enter file name with '.csv' (e.g., example.csv):

File saved successfully to test.csv

Press any key to continue...

As you can see we change middle cell than E1 E2 E3 changed

Cell [E,3]: =STDDEV(A1..C3)

	A	B	C	D	E	F	G	H	I	J
1	1	2	3		45.000000					
2	4	5	6		5.000000					
3	7	8	9		2.581989					
4										
5										
6										
7										
8										
9										
10										

Welcome Excell Simulation!

- 1.) Move with Arrow Keys. Select cell for press Enter
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- 3.) You can select multiple cells. =AVER(A1..C4)
- 4.) Save to file press 's'
- 5.) Load from file press 'l'
- 6.) Exit the program for press 'q'

C

Cells expands and labels also expands to AA AB .. ZZ and 1 2 3 ... 99

Cell [A,1]:

	W	X	Y	Z	AA	AB	AC	AD	AE	AF
15										
16										
17										
18										
19										
20										
21										
22										
23										
24										

Welcome Excell Simulation!

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- 5.) Load from file press 'l'
- 6.) Exit the program for press 'q'

Cell [C,1]: =A1+B1

	A	B	C	D	E	F	G	H	I	J
1	2	4	6.000000							
2										
3										
4										
5										
6										
7										
8										
9										
10										

Welcome Excell Simulation!

- 1.) Move with Arrow Keys. Select cell for press Enter
- 2.) You can calculate two cells. =A1+B1
- 3.) You can select multiple cells. =AVER(A1..C4)
- 4.) Save to file press 's'
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